Adaptive Probe-Guided Genetic Algorithm (APGGS) — Experiment Report

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Engine Version: v4.2

1. Executive Summary & Dataset Metadata

Dataset Name	DEMO10	**Cities / Nodes**	10
Edge Weight Type	EUC_2D	**Known Best Optimum**	Unknown
Final Global Best Cost	102.00	**Optimality Gap**	Unavailable
Best Found Generation	0	**Completed Generations**	30
Stagnation Streak	30 / 30	**Stopping Condition**	30 gens without improvement
Experiment Status	STOPPED_STAGNATION	**Total Execution Time**	0.12 s
Stop Reason	No improvement for 30 consecutive generations	**Random Seed**	1

2. Initial High-Quality Route Construction

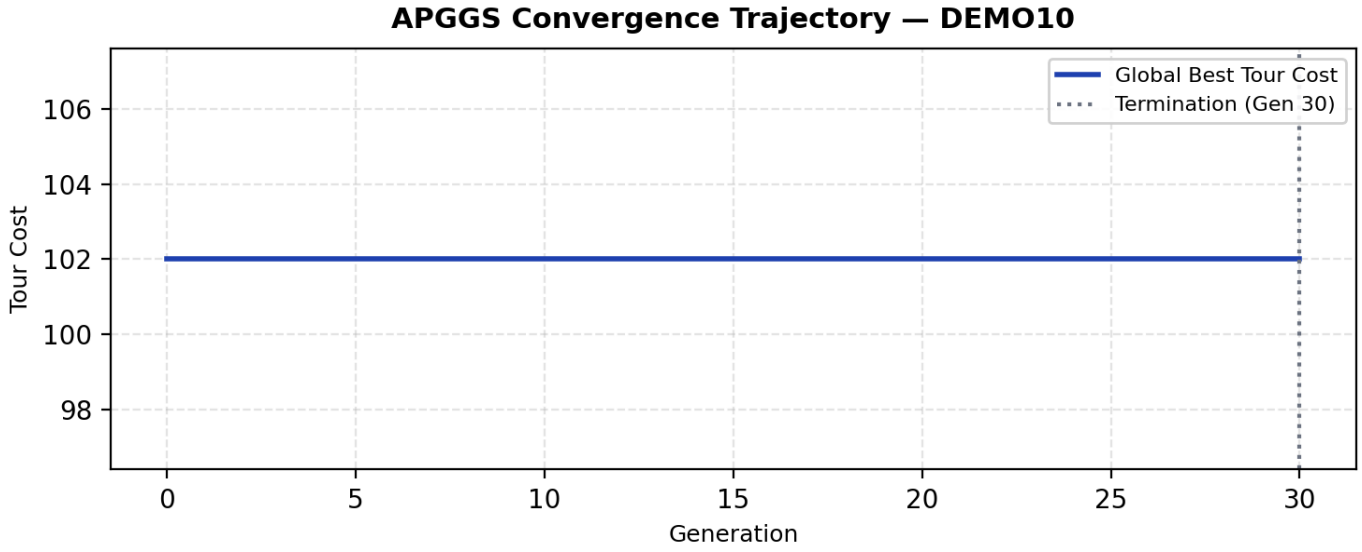
Construction Stage	Resulting Tour Cost	Improvement vs Prev	Duration (s)
Stage 1: Greedy Nearest-Neighbor Tour	105.00	Baseline	0.0000
Stage 2: 2-Opt Local Search Improvement	102.00	-3.00	0.0000
Stage 3: 2-Opt Deep Refinement Seed	102.00	-0.00	0.0000
Stage 4: Initial 70-Individual Population (NN+Greedy+Random)	102.00	Population Seeded	0.0005

3. Genetic Algorithm & Adaptive Pipeline Configuration

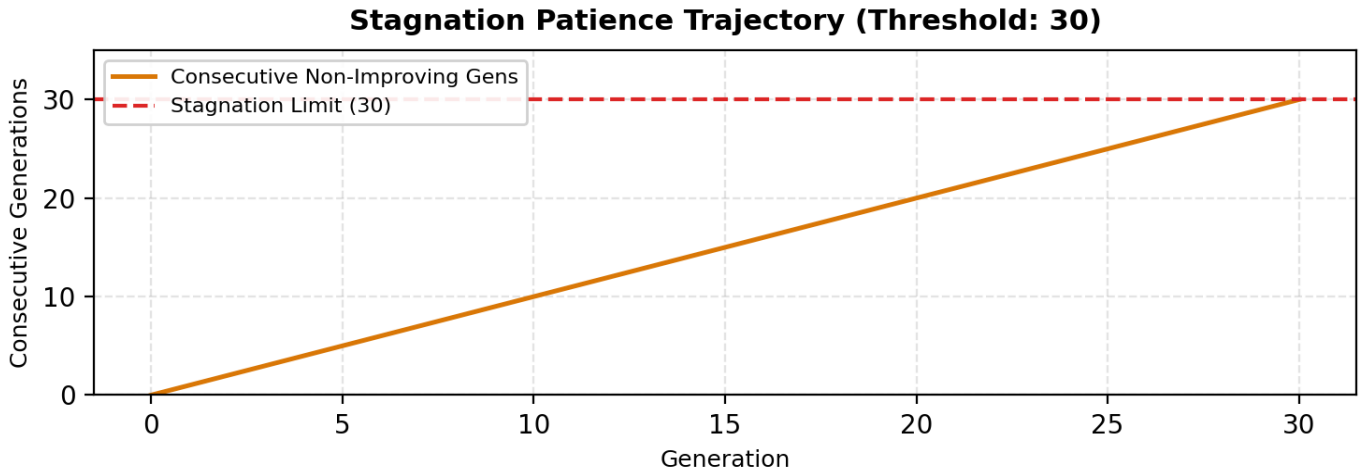
Parameter	Configured Value	Parameter	Configured Value
Population Size	30	Mutation Rate	0.20
Nearest-Neighbor Seeds	17	Swap Fraction	0.02
Greedy Edge Seeds	11	Segment Crossover Prob	0.40
Random Tour Seeds	2	Offspring 2-Opt Prob	0.35
K-Nearest Neighbors	15	Adaptive Elites Count	14
Tournament Size	4	PMG Memory Decay	0.92
Selective LKH Cadence	Every 15 gens	Stagnation Window	15
Selective LKH Individuals	3	Stagnation Patience Limit	30 consecutive gens

4. Research Visualizations & Trajectory Curves

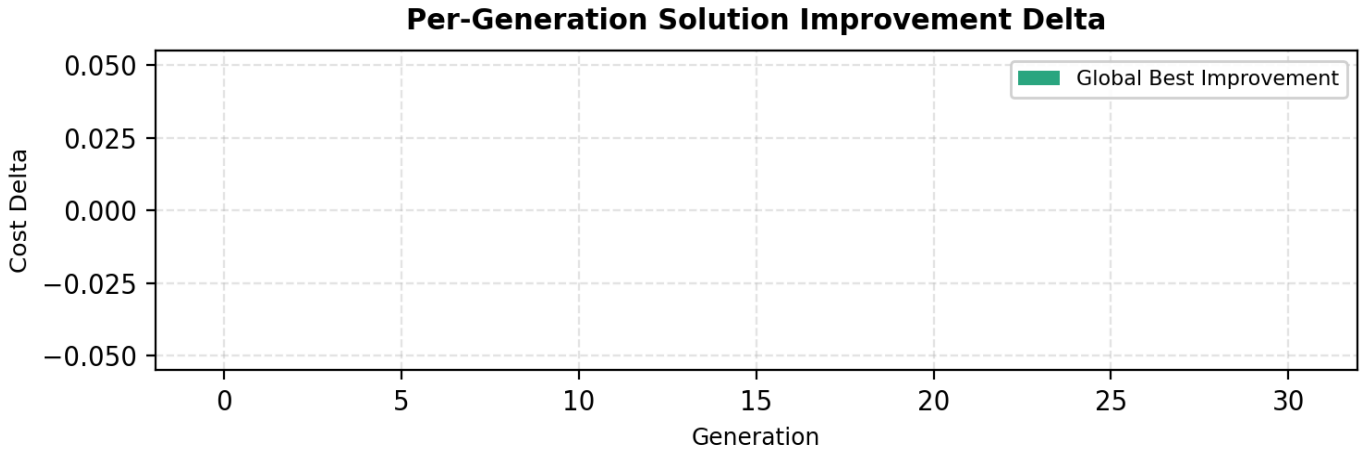
Convergence Trajectory: Global Best Cost vs Generation

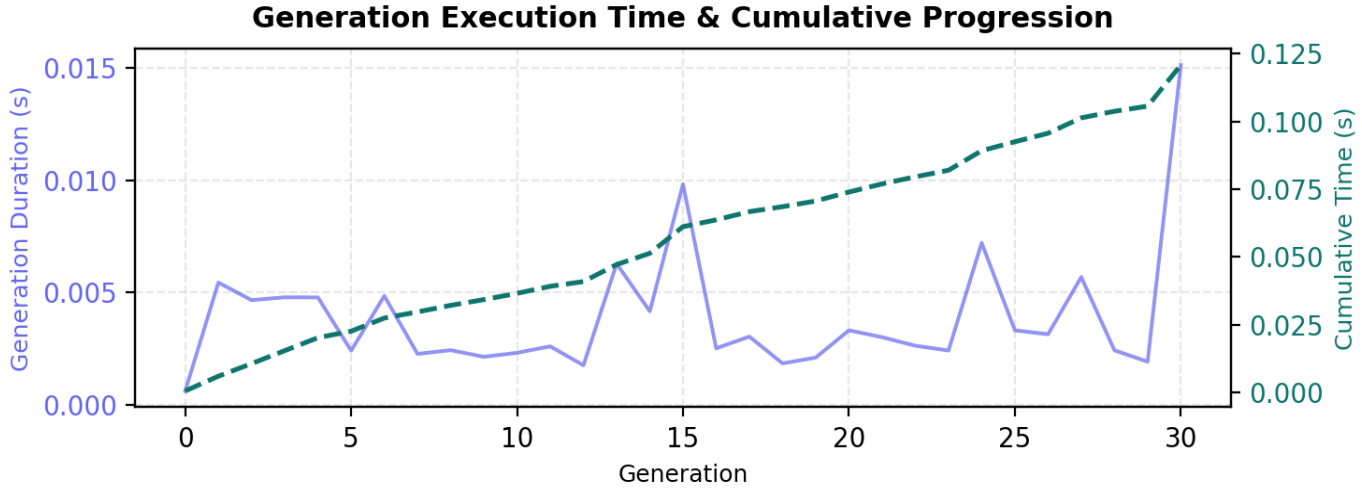


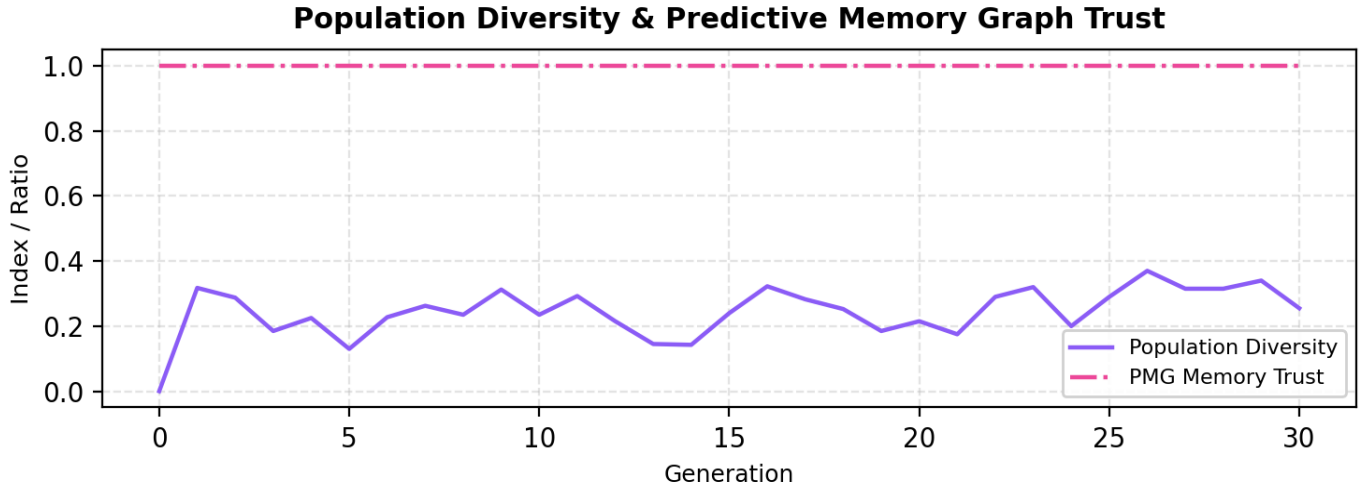
Stagnation Counter Progression (Target: 30 Consecutive Non-Improving Gens)



Per-Generation Cost Improvement & Runtime Progression







5. Evolution Highlights (Global-Best Discoveries)

Gen	Global Best Cost	Cost Reduction	Gap (%)	Operator / Method	PMG Trust	Non-Imp Streak	Duration (s)
0	102.00	0.00	—	Nearest Neighbor (start=0)	1.00	0	0.0006
End: 30	102.00	0.00	—	Terminated (No improvement for 30 consecutive generations)	1.00	30	0.0151

6. Final Discovered Tour & Solution Vector

Optimized Tour Length: 102.00 | **Optimality Gap:** Unavailable | **Discovered at Generation:** 0

Permutation Sequence Preview:

`0 → 7 → 8 → 1 → 9 → 2 → 3 → 4 → 5 → 6`

7. Execution Event Log & Diagnostics

- `Initialized demo10 (10 cities). Initial best cost: 102.00`
- `Gen 30: Stagnation limit reached (30 non-improving generations). Terminated.`

8. Artifact Manifest & Research Reproducibility

All raw experimental records have been saved to runs/demo10_baseline/ including summary.json, generation_history.jsonl, best_route.csv, and report.pdf.